

# Wael Bahgat

## Principal Software Engineer

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### SUMMARY

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Principal software engineer with 15+ years of C++ application and server-side software for satellite communications, aviation surveillance, maritime autonomy and telecom. Designs frameworks with a stable core that other engineers extend without changing shipped code; puts in place the simulation and release pipelines that let teams test every day and ship what was tested; leads delivery from customer requirements to production.

### HIGHLIGHTS

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- Designed and built from the first line a surveillance system framework with a stable core that other engineers extend without changing it; the approach was adopted across an aviation surveillance department. Has since applied it to maritime autonomy and a satellite core network.
- Twice replaced slow, costly testing on real systems with simulation: a playback simulator for an autonomous vessel, and software-in-the-loop simulation for satellite ground software.
- Led a team of five engineers through a full delivery cycle, from customer requirements to deployment; mentors engineers and takes part in hiring.

### CORE SKILLS

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**Languages:** C++, Qt, Java, Python (tooling and automation), SQL and PL/SQL, Dart/Flutter

**Architecture:** Object-oriented design, plugin frameworks, SOLID and the open/closed principle, design patterns, multithreading, thread and session pools, REST and TCP services, interface definitions between systems

**Testing:** GTest, GCov, test-driven development, software-in-the-loop and playback simulation, performance and load testing

**Delivery:** Docker, GitLab CI/CD, release engineering, amd64/arm64 cross-compilation, Linux, systemd, gdb

**Data and cloud:** PostgreSQL, MongoDB, Oracle, SQLite; AWS (ECS Fargate, RDS, S3, CloudFront)

### EXPERIENCE

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#### Principal Software Engineer, Lead Developer, Core Network

Dec 2023 – Present

**Elveo Mobile** · *Direct-to-device satellite operator connecting ordinary mobile phones to satellites.*

- Co-designed and built the plugin architecture of the dual-mode (satellite and ground) GSM/LTE core network software in C++: features are separately loaded modules around a stable core, so capability is added without changing shipped code. Have led its development since.
- Designed the framework's core libraries: per-module configuration, asynchronous thread-pool processing, a generic database session pool, and interchangeable back ends (PostgreSQL, MongoDB, SQLite).
- Created the container build system and a development environment produced from the same definitions as production, for amd64 and arm64; adopted as the team's standard way of developing.
- Designed and own the GitLab CI/CD release pipeline: unit tests and coverage, multi-stage Docker builds, release candidates, and promotion of the tested candidate to release.
- Built internal operator tooling (web front end, REST back end, PostgreSQL, Python command-line tool) and introduced software-in-the-loop simulation for testing; mentored an intern through its implementation to delivery.
- Integrated GTest and GCov across all components; take part in the team's hiring interviews.

#### Senior Software Specialist

Sep 2021 – Dec 2023

**L3Harris** · *Aviation surveillance.*

- Built core C++ components for aviation surveillance systems.

- Designed a framework for multiple service applications sharing a common core with extendable layers: configuration manager, log manager, thread pool, pluggable business logic and XML-driven configuration.
- Applied test-driven development with GTest, coverage analysis with GCov, and performance testing with custom profilers.

### Development Team Lead

Jul 2020 – Sep 2021

**Gibbs & Cox, a Leidos company** · *Proof-of-concept phase of an autonomous navigation system for surface vessels.*

- Learned the maritime domain from the existing code and from the vessel itself, and led the team of five software engineers within a year: customer requirements, architecture, implementation, testing and deployment.
- Designed a modular Qt/C++ framework that models onboard subsystems as generic components behind a hardware abstraction, so the software is not tied to one vessel.
- Built a playback simulator from recorded data, so use cases run at a desk instead of at sea, and an operator interface for development and testing.

### Senior Software Specialist

May 2016 – Jul 2020

**L3Harris** · *Aviation surveillance.*

- Aggregated radar and ADS-B data in real time across multiple sites, and developed Qt/C++ air traffic display applications.
- Designed and built a new surveillance system from scratch on a framework of decoupled components with a thread pool, central logging and configuration-driven behavior; the approach was later adopted across the department's projects.
- Built a drone surveillance user interface in C++/Qt, and reusable C++ libraries with GTest coverage.
- Member of the team that developed a voice switching solution for air traffic control, supporting hybrid analog and VoIP calls.

### Lead Software Engineer, Production Engineering

May 2012 – Apr 2016

**Openet Telecom**

- Delivered hotfixes and patches across the production stack in C++ and Java for carriers including Verizon, AT&T and Sprint.
- Diagnosed memory leaks, deadlocks and crashes with gdb and mdb, and took critical customer issues from first report to delivered patch.

### Software Engineer to Lead Software Engineer, Professional Services

Jan 2005 – May 2012

**Openet Telecom**

- Designed and delivered telecom mediation and billing solutions for major carriers in Java and PL/SQL on high-volume Oracle databases.
- Authored requirements, performance test specifications and architecture documents.

## INDEPENDENT PROJECT

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### Yalabeena, a marketplace for booking sessions with hosts (yalabeena.com)

- Designed, built and operate the whole product alone: one Flutter code base for web, Android, iOS and desktop in English and Arabic, a C++ REST back end built from plugin modules, and PostgreSQL. Includes payments, in-app video sessions and chat, push notifications and an assistant.
- More than 200 end-to-end test files across four platforms, and a release-candidate process that proves each release on the web before it reaches the app stores.

## EDUCATION

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**M.S. Electrical Engineering** · NYU Tandon School of Engineering, Brooklyn, NY

GPA 3.87

**B.S. Computer Engineering** · Ain Shams University, Cairo, Egypt